Practice workbook · The quotient rule: differentiating fractions

Differentiating a fraction u/v, simplifying the result and using it to give a gradient or to show the gradient always has one sign.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-02

1. Question 10 · part d · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part d only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

1. Question 10 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part a · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

8 Given that $2xy + 5y = e^x$

(a) show that $\frac{dy}{dx} = \frac{y(2x+3)}{2x+5}$

(5)

(b) find the value of $\frac{dy}{dx}$ when $x = 0$

(2)

(c) find an equation of the normal to the curve with equation $2xy + 5y = e^x$ at the point where $x = 0$
Give your answer in the form $px + qy + r = 0$ where p, q and r are integers.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

P 6 2 2 8 0 A 0 1 8 3 2

2. Question 8 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

8 Given that $2xy + 5y = e^x$

(a) show that $\frac{dy}{dx} = \frac{y(2x+3)}{2x+5}$

(5)

(b) find the value of $\frac{dy}{dx}$ when $x = 0$

(2)

(c) find an equation of the normal to the curve with equation $2xy + 5y = e^x$ at the point where $x = 0$
Give your answer in the form $px + qy + r = 0$ where p, q and r are integers.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

P 6 2 2 8 0 A 0 1 8 3 2

3. Question 8 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

3 Differentiate with respect to x

(a) $e^{2x}\sqrt{5x-3}$

(3)

(b) $\dfrac{x^3}{\cos 3x}$

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 6 3 0 7 A 0 6 3 2

4. Question 3 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 3 is 6 marks)

4. Question 3 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part whole · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

8 Use calculus to find the gradient of the curve with equation

$$y = e^{3x}(9x+2)^{\frac{1}{3}}$$

at the point on the curve where $x = 1$
Give your answer to 3 significant figures.

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 6 5 0 9 A 0 2 2 3 2

5. Question 8 · part whole · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 8 is 5 marks)

5. Question 8 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 10 · part d · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

6. Question 10 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 7 · part b · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

7 The curve C has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on C.

(5)

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

(4)

(c) Hence, or otherwise, determine the nature of each of these stationary points.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 6 0 2 4 A 0 1 8 3 6

7. Question 7 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 5 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 Differentiate with respect to x

(a) $e^{4x}(6x+2)^{\frac{3}{2}}$

Give your answer in the form $e^{4x}\left(\sqrt{6x+2}\right)(Ax+B)$ where A and B are integers.

(5)

(b) $\dfrac{\sin 3x}{(2x-4)^3}$

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

8. Question 5 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 5 is 8 marks)

8. Question 5 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 2 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

2 $y = (\sin 2x)\sqrt{3 + 2x}$

Show that $\frac{dy}{dx} = \frac{\sin 2x + (A + Bx)\cos 2x}{\sqrt{3 + 2x}}$ where A and B are integers to be found. **(5)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 2 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 2 continued

OLevels.com SCHOOL

(Total for Question 2 is 5 marks)

9. Question 2 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 4 · part whole · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

4 $y = \dfrac{\sin 2x}{\sqrt{x^2 - 9}} \qquad |x| > 3$

Show that $\dfrac{dy}{dx} = \dfrac{2(x^2 - 9)\cos 2x - x\sin 2x}{\sqrt{(x^2 - 9)^3}}$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 4 · part whole · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part whole only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 4 is 5 marks)

10. Question 4 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 10 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

30

P 7 1 6 4 2 A 0 3 0 3 6

11. Question 10 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part d · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

24

P 7 4 2 8 4 A 0 2 4 3 2

12. Question 9 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 11 · part a · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

13. Question 11 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 3 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

3 The curve C has equation $y = e^{3x}(2x-1)^4$

Using calculus, find the exact value of the gradient of the tangent to C when $x = 1$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 3 5 8 7 A 0 6 3 2

14. Question 3 · part whole · 5 marks

2024 NOV · Paper 02

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

OLevels.com SCHOOL

(Total for Question 3 is 5 marks)

14. Question 3 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 6 · part a · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 The curve C has equation $y = \dfrac{e^{(x^2+1)}}{x^2+1}$

(a) Show that $\dfrac{dy}{dx} = \dfrac{Kx^3e^{(x^2+1)}}{(x^2+1)^2}$ where K is a constant whose value is to be found.

(5)

(b) Find an equation of the tangent to C at the point on C where $x = -1$
Simplify your answer.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

15. Question 6 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 8 · part a · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

8 $y = \dfrac{2e^{3x+1}}{5x^2}$

(a) Find $\dfrac{dy}{dx}$

Give your answer in the form $\dfrac{Ae^{3x+1}(Bx-A)}{Cx^3}$ where A, B and C are prime numbers to be found.

(5)

The value of x increases by 2%

(b) Use your answer to part (a) to find an estimate, in terms of x, for the percentage change in y

Give your answer in the form $(Px-Q)$ where P and Q are integers.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 8 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL